KYB — ERP Integrator & Administrator Technical Manual

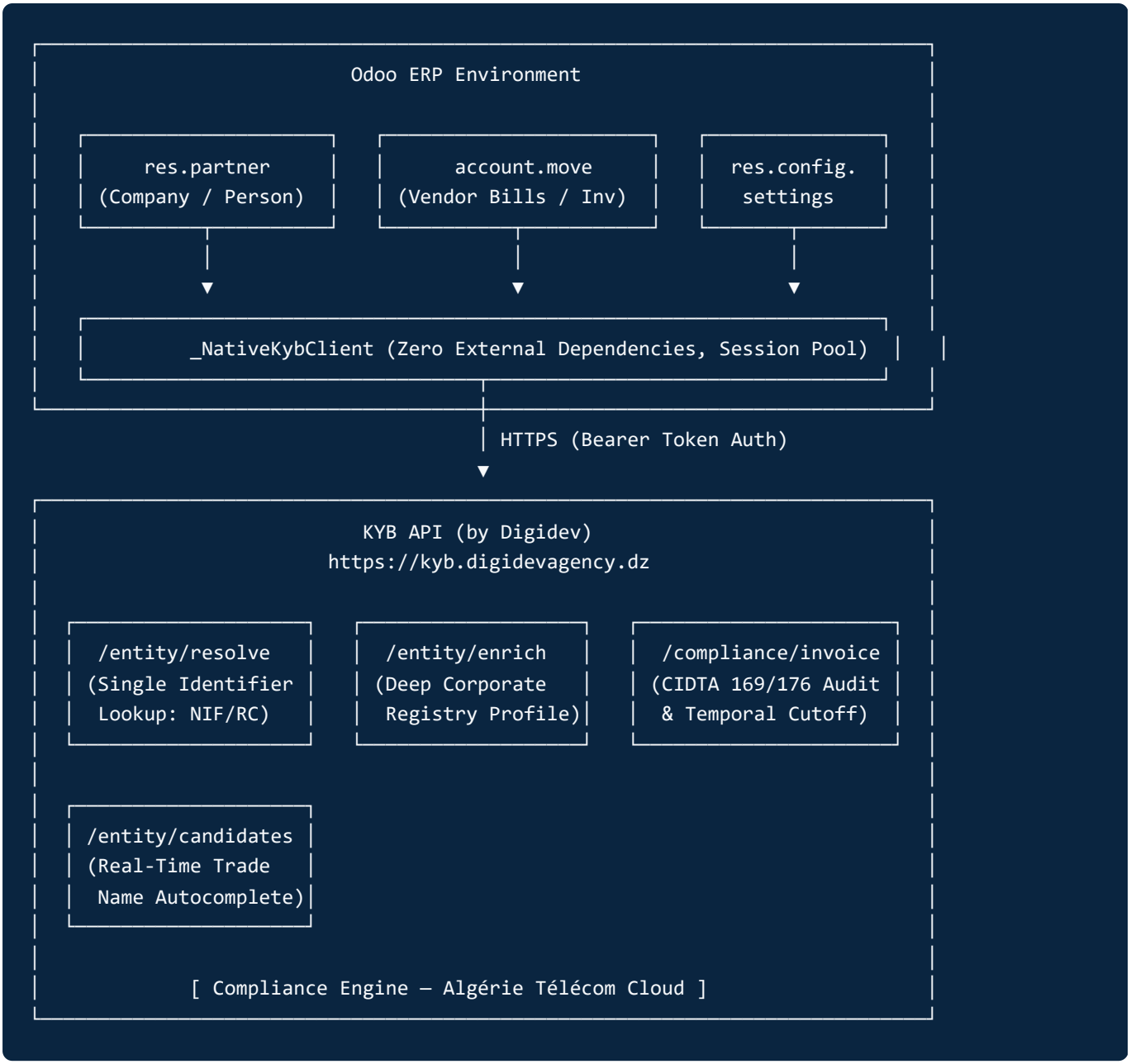
Version: `1.4.2` / `20.0.1.4.2` **Platform Domain:** kyb.digidevagency.dz **Administration Portal:** kyb.digidevagency.dz **Target Environments:** Odoo 16.0 LTS, 17.0, 18.0, 19.0, 20.0 (Community & Enterprise) **Legal Framework:** Algerian Law 25-11 (Data Residency), Law 18-07 (Personal Data Protection), Décret Exécutif 05-468 & 18-112 (RC Invoicing Standard), CIDTA Art. 169 & 176, CTCA Art. 30 (Temporal Invoice Compliance).

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1. Executive Summary & Architecture

The **KYB Odoo Connector** (`connectors/kyb_odoo`) is an institutional-grade Model/Record (MR) bridge connecting Odoo ERP installations directly to the **KYB API Gateway (by Digidev)**.

Hosted exclusively within Algerian national borders on the **Algérie Télécom Cloud** (`kyb.digidevagency.dz`), the gateway provides sub-second corporate identity resolution, real-time national trade registry synchronization, automated fiscal duplicate detection, and temporal invoice tax compliance verification.



Core Architecture Invariants

1. **Thin Connector / Thick Core Architecture:** All regulatory classification, trade registry synchronizations, and temporal status computations live inside the KYB API (by Digidev). The Odoo connector contains zero tax liability assertions and operates strictly as a secure RPC data consumer.
2. **Zero External Python Dependencies:** Built entirely with standard Python `requests` bundled natively with Odoo. No offshore pip packages or compiler toolchains required.
3. **Session Pooling & Keep-Alive:** Maintains a process-level client pool (`_CLIENT_CACHE`) with TCP/TLS connection reuse, delivering sub-30ms response times on cached lookups.
4. **Sovereign Data Residency (Loi 25-11):** 100% of API endpoints and sovereign engines reside strictly within Algerian data centers. Zero PII leaves Algerian sovereign territory.

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2. Customer Admin Portal: Credential Governance, Sub-Keys & Subscription Tiers

Enterprise administrators manage their KYB subscription, credentials, and consumption via the **KYB Customer Administration Portal** at kyb.digidevagency.dz.

A. Master Tenant Pool vs. Scoped Sub-Keys

- **Master Tenant Account:** Represents the enterprise contracting entity. Billing credits and master subscriptions are centralized at the tenant level.
- **Sub-Keys (`sk_live_...` / `sk_test_...`):** Downstream ERP connectors authenticate exclusively via scoped Sub-Keys. Each Sub-Key is linked to the tenant's shared master credit pool while maintaining independent metrics and security constraints.

[Image: Tenant Credit Pool Management]

Figure 1: Master Tenant Credit Pool Provisioning & Enterprise Plan Cockpit.

[Image: Customer Portal Client Overview]

Figure 2: Enterprise Client Dashboard showing Master Pool balance and 14-day API consumption velocity.

B. Subscription Tiers & Sub-Key Quotas

KYB applies strict, transparent credential governance based on the organization's subscription tier:

Subscription Tier	Master Credit Pool	Max Allocated Sub-Keys	Target Deployment Topology
Starter	1,000 / month	Up to **3 Sub-Keys**	Single Odoo instance (Production + Staging sandbox).
Growth	10,000 / month	Up to **10 Sub-Keys**	Multi-branch Odoo deployments (HQ, Regional branches, UAT).
Enterprise	50,000+ / month	**Custom / Unlimited**	Distributed multi-company conglomerates with automated CI/CD.

[Image: Sub-Keys Management Cockpit]

Figure 3: Scoped Sub-Keys Management showing allocated limits, monthly usage counters, and environment scoping.

C. Portal Sub-Key Lifecycle & Governance

1. **Capacity Meter:** The portal dashboard displays real-time Sub-Key utilization (`Active Sub-Keys` vs. `Allocated by Subscription`). If the quota is reached, generating additional sub-keys requires archiving an unused key or upgrading the subscription tier.
2. **Dedicated Environment Partitioning:** Administrators must create separate sub-keys for production ERP environments and development/staging environments (`is_sandbox = True`). Sandbox queries (`0000...` magic identifiers) never consume credits.
3. **Usage Attribution & Chargeback:** The customer portal records consumption on a per-sub-key basis. Enterprises can attribute monthly verification volumes to specific subsidiary ERP installations.
4. **Network IP Whitelisting:** Administrators can restrict each sub-key to authorized outbound public IP addresses (`allowed_ips`). Any request originating outside the configured IP block is rejected with HTTP 403.
5. **Instant Revocation & Rotation:** If a server is decommissioned or an API key is compromised, clicking **Revoke** in the customer portal immediately invalidates the key across the gateway in sub-milliseconds without affecting any other operational ERP systems.

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3. Module Lifecycle: Installation, Upgrades & Configuration

A. Module Discovery & Installation in Apps Catalog

1. Navigate to **Apps** in Odoo.
2. In the search box, search for `kyb` (clear the default `Apps` category filter to view institutional localization packages).
3. The official **KYB — Algerian Corporate Identity & Temporal Fiscal Compliance** card appears.

[Image: Apps Catalog Search]

Figure 4: Discovering the KYB connector in the Odoo Apps catalog.

B. Module Details & Administrative Actions (Upgrade / Uninstall)

Clicking the kebab menu (three vertical dots) on the module card exposes all administrative actions:

- **Info sur le module (Module Details):** Displays full technical details, version code (`20.0.1.4.1`), author ("Mouadh Benhassine / Digidev Agency"), license, and dependencies (`account` , `contacts`).
- **Mettre à niveau (Upgrade):** Forces reloading of all XML views, Python models, and security rules without server downtime.
- **Désinstaller (Uninstall):** Clean uninstallation removing database views while preserving standard Odoo partner data.

[Image: Module Actions Menu]

Figure 5: Module actions dropdown showing Module Info, Upgrade, and Clean Uninstallation.

[Image: Module Installed Technical Form]

Figure 6: Official module technical sheet detailing version 20.0.1.4.1 and dependencies.

C. System Parameters Configuration (`res.config.settings`)

Navigate to **Invoicing (or Contacts) → Configuration → Settings → KYB Sovereign Verification**:

Parameter Key	Default Value	Description
`kyb.api_key`	*None*	Sovereign Sub-Key (`sk_live_...` for production or `sk_test_...` for testing) generated from the Customer Portal.
`kyb.base_url`	`https://kyb.digidevagency.dz`	Ingress endpoint of the KYB API Gateway (by Digidev).
`kyb.timeout`	`35.0`	Client HTTP timeout in seconds (aligned with Gateway 35s SLA).
`kyb.sovereign_autocomplete`	`True`	Intercepts company name typing to query the National Trade Register in real time.
`kyb.allow_worldwide_fallback`	`False`	Allows fallback to international search when explicitly clicking *Recherche mondiale* 🌍.
`kyb.duplicate_nif_policy`	`warn`	Fiscal duplicate prevention policy: `warn` (modal notification), `block` (raises UserError), or `ignore`.
`kyb.invoice_audit_enabled`	`True`	Activates temporal vendor bill audit against trade registry status at invoice date.
`kyb.invoice_blocking_policy`	`strict_block`	Invoicing validation policy: `strict_block` (interdit comptabilisation) or `soft_warn` (warning in chatter).

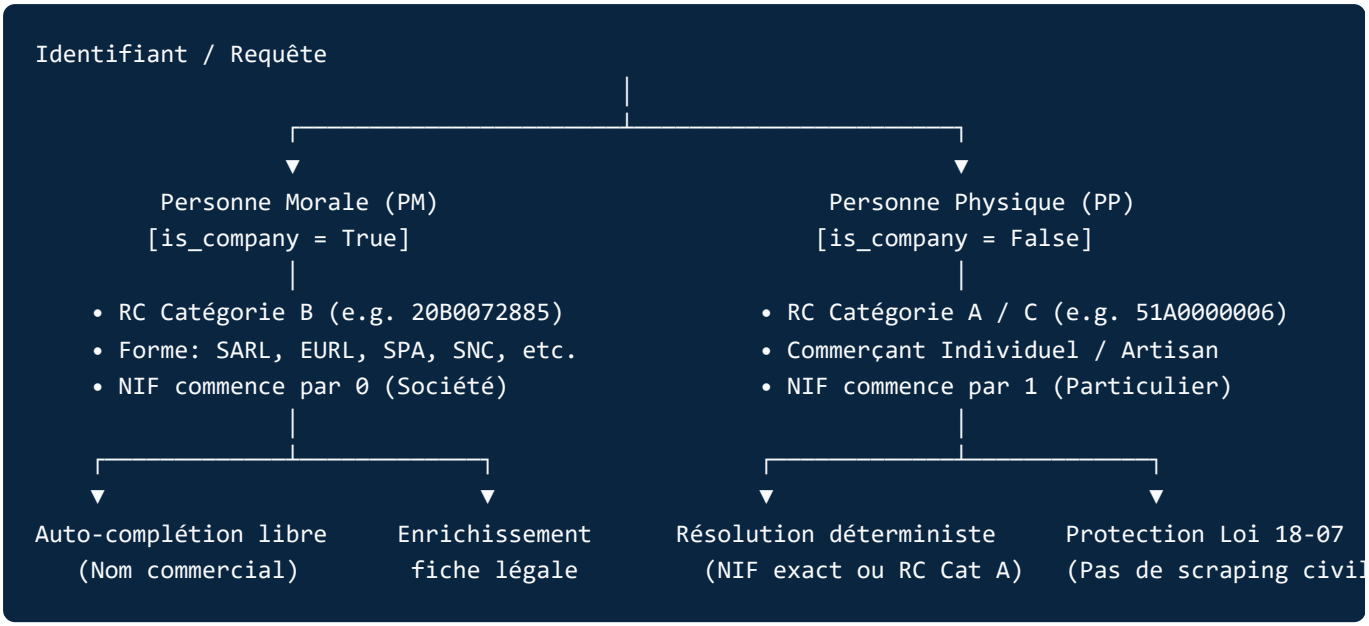
[Image: KYB Settings Configuration]

Figure 7: Complete KYB Sovereign Verification settings panel in Odoo 20.

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4. Algerian Corporate Identity Partitioning: Company vs. Person Contacts

Under **Algerian Law 18-07** (*Protection des personnes physiques dans le traitement des données à caractère personnel*) and corporate privacy governance, identity resolution is strictly partitioned:



A. Company Contacts (`Personne Morale`, `is_company = True`)

- **Commercial Registry (RC):** Category `B` (e.g., `20/00-0072885B01`).
- **Legal Form:** SARL, EURL, SPA, SNC, SCS, SCA, GIE, EPE.
- **Tax ID (NIF):** 15-digit national fiscal number starting with `0`.
- **Search Autocomplete:** Open fuzzy discovery by company name (`autocomplete_by_name`) is fully enabled. Typing a partial name in Odoo partner dropdowns instantly queries the sovereign index.

[Image: Verified Company Partner]

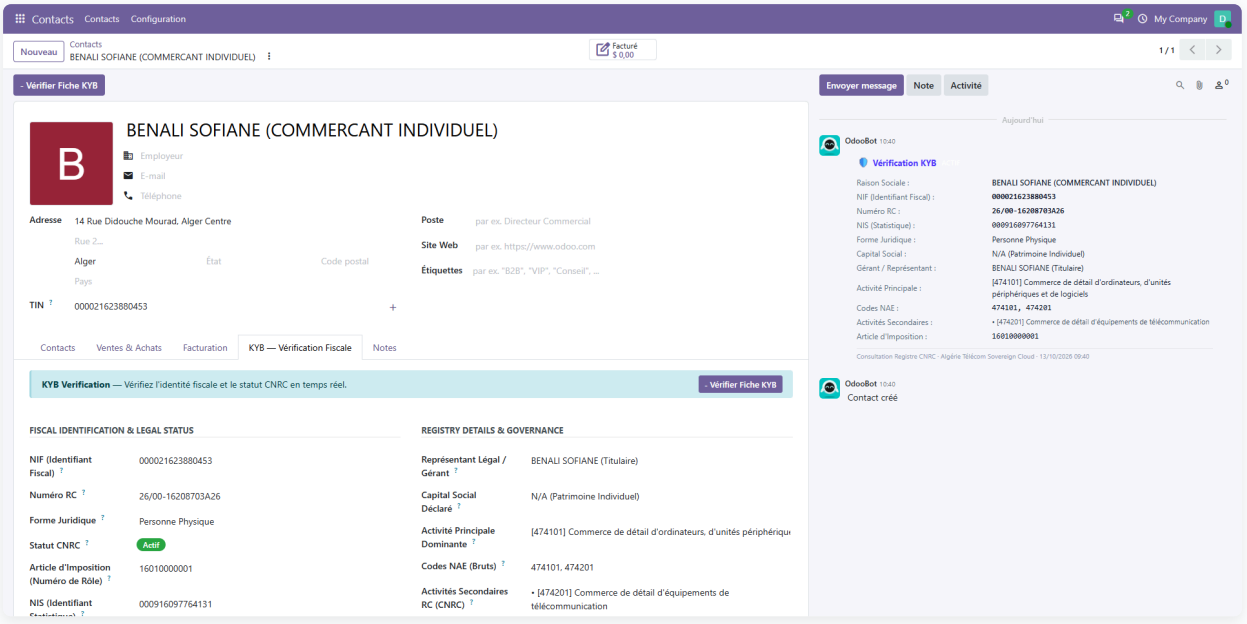
Figure 8: Fully enriched Company Partner (DIGI DEV AGENCY, EURL) displaying canonical Décret 05-468 RC format (20/00-0072885B01) and green ACTIF badge.

[Image: Wilaya 00 Legacy National Partner]

Figure 9: Legacy National Commercial Register entity (BELLAT FER DISTRIBUTION BFD) formatted to canonical 00/00-0012637B00 (National Commercial Register Standard).

B. Person Contacts (`Personne Physique` , `is_company = False`)

- **Commercial Registry (RC):** Category `A` (Commerçant individuel) or Category `C` (Artisan / Coopérative) (e.g., `51/00-0000006A` , `16/00-0012345C16`).
- **Legal Form:** Automatically tagged as `"Personne Physique"` .
- **Tax ID (NIF):** 15-digit fiscal identifier starting with `1` .
- **Privacy Boundary (Law 18-07):** Natural citizens' trade registers contain private biographical data (civil name, date/place of birth, home address). Speculative free-text civil name scraping is strictly blocked at the gateway level.
- **Resolution Flow:** Integrators and users resolve natural persons **deterministically** by providing the official **15-digit NIF** or **Category A/C RC**.



Verified Person Partner

Figure 10: Enriched Personne Physique contact (BENALI SOFIANE, Commerçant) classified with is_company=False, Category A RC (26/00-16208703A26).

[Image: Verified Artisan Partner]

Figure 11: Enriched Artisan contact (COOPÉRATIVE ARTISANALE DU BOIS) classified with Category C RC (16/00-0012345C16).

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5. Fiscal Duplicate Detection & Native Deduplication

Why Duplicate Contacts Occur in Standard Odoo

In native Odoo, `res.partner` intentionally has **no unique constraint** on `name` or `vat`. Employees entering contacts from sales, invoicing, or purchasing channels frequently create duplicate partner cards.

KYB Proactive Duplicate Prevention

Under Décret 05-468, the Algerian **NIF** is a strictly unique sovereign identifier. KYB provides a three-tiered protection layer:

1. **Autocomplete Existence Badging:** When typing a company name in the autocomplete dropdown, if an entity already exists in the local database, the suggestion displays:

`CONSTANTINE • RC: 20/00-0072885B 🟢 Actif • ⚠️ Existant (#15)`

1. **Onchange Fiscal Guard (`kyb_duplicate_nif_policy`):**

When resolving an entity or saving a NIF: - If `kyb_duplicate_nif_policy = 'warn'` (default): Displays an institutional warning alert directing the user to the existing contact ID. - If `kyb_duplicate_nif_policy = 'block'`: Raises a strict `UserError` preventing duplicate creation.

1. **Native Odoo Merge Wizard Integration:**

Administrators can consolidate existing duplicate cards using Odoo's native wizard: **Contacts → Select Records → Actions → Fusionner (Merge)**. The connector guarantees that all canonical KYB fields (`kyb_nif`, `kyb_rc`, `kyb_statut_cnrc`, `kyb_forme_juridique`) are preserved on the master destination record.

[Image: Real-Time Autocomplete Dropdown]

Figure 12: Real-time company name autocomplete dropdown querying live national trade registry.

[Image: Duplicate NIF Alert Modal]

Figure 13: Institutional warning alert displayed when entering an already-registered NIF, referencing partner ID 15.

[Image: Native Deduplication Merge Wizard]

Figure 14: Native Odoo Partner Merge Wizard (base.partner.merge.automatic.wizard) consolidating duplicate records while preserving KYB audit data.

6. Temporal Invoice Compliance (CIDTA Art. 169 & 176 / CTCA Art. 30)

Under Algerian Tax Law, expenses and VAT originating from suppliers deregistered (radié) from the Trade Register at the invoice emission date are strictly **non-deductible**.

Verification Workflow

1. When a vendor bill (`account.move` , `move_type = 'in_invoice'`) is created or confirmed, the module evaluates the supplier's standing at `invoice_date` .
2. If `invoice_date > kyb_date_radiation` :

- `kyb_compliance_status` transitions to `non_compliant` . - A critical red banner is displayed on the invoice form. - If `kyb_invoice_blocking_policy = 'strict_block'` , confirmation is rejected with an explanatory audit notice.

[Image: Vendor Bills Compliance Tree]

Figure 15: General Vendor Bills list view displaying real-time Conformité Fiscale KYB status badges across all statutory scenarios.

[Image: Compliant Vendor Bill]

Figure 16: Compliant vendor bill (FACT-DIGI-2026-089) from active supplier: tax_deductible=True.

[Image: Non-Compliant Vendor Bill]

Figure 17: Critical red danger alert banner on bill from struck-off vendor (SARL ANCIENNE DISSOLTE, Radié 15/06/2021).

[Image: Cautionary Warning Vendor Bill]

Figure 18: Cautionary warning on historical bill (FACT-HISTORIQUE-2020-012) dated prior to vendor strike-off.

[Image: Risk Invoices Cockpit View]

Figure 19: Dedicated KYB Risk Cockpit isolating all invoices presenting statutory tax risks.

7. Data Dictionary (`res.partner` & `account.move`)

`res.partner` Extended Fields

Field Technical Name	Type	Description	Legal Reference
`kyb_nif`	`Char(20)`	15-digit National Fiscal Identifier	DGI / Décret 05-468
`kyb_rc`	`Char(30)`	Official Commercial Register (`WW/00-XXXXXXX[A-D]YY`)	Décret 05-468 & 18-112
`kyb_nis`	`Char(30)`	Statistical Identifier (ONS)	ONS
`kyb_article_imposition`	`Char(30)`	Article d'Imposition (Numéro de Rôle)	Secret Fiscal (Art. 114 CPF)
`kyb_statut_cnrc`	`Selection`	Status: `ACTIF` , `RADIE` , `SUSPENDU` , `INCONNU`	CNRC Sidjilcom
`kyb_date_radiation`	`Date`	Official deregistration date	CNRC BOAL Gazette
`kyb_forme_juridique`	`Char`	Legal structure (`SARL` , `EURL` , `Personne Physique` ...)	Code de Commerce
`kyb_capital_social`	`Char`	Registered Share Capital (e.g. `10 000 000 DZD`)	CNRC
`kyb_representant_legal`	`Char`	Statutory Legal Representative / Gérant	CNRC Statuts
`kyb_activite`	`Char`	Dominant Primary Activity (with NAE code)	Nomenclature NAE
`kyb_nae_codes`	`Char`	Comma-separated official NAE codes	CNRC
`kyb_activites_rc`	`Text`	Full catalog of secondary registered activities	CNRC
`kyb_last_verified`	`Datetime`	Timestamp of last successful API resolution	Audit Trail

`account.move` Extended Fields

Field Technical Name	Type	Description
`kyb_compliance_status`	`Selection`	`compliant` , `non_compliant` , `warning` , `unverified`
`kyb_tax_deductible`	`Boolean`	Computed tax deductibility indicator
`kyb_audit_date`	`Datetime`	Timestamp of invoice tax compliance verification
`kyb_compliance_message`	`Text`	Neutral regulatory compliance notice

8. Operational Scenarios & Verification Matrix (18/18 Scenarios PASS)

The connector's resilience, compliance, and edge-case handling have been verified via an automated end-to-end verification test suite executed on the production runtime. All 18 scenarios achieved a **100% Pass Rate**:

#	Code	Scenario Description	Tested Identifier / Action	Verified Behavior & Screenshot Reference	Status
01	`S01`	**RC Category B (Corporate)**	`20/00-0072885B01`	Full two-phase resolution: EURL DIGI DEV AGENCY, `is_company=True`, Status `ACTIF` (*Figure 8*).	🟢 PASS
02	`S02`	**RC Category A (Personne Physique)**	`26/00-16208703A26`	Deterministic resolution: BENALI SOFIANE, `is_company=False`, Forme `Personne Physique` (*Figure 10*).	🟢 PASS
03	`S03`	**RC Category C (Artisan / Coop)**	`16C0012345`	Native Python parser extracts category `C` and formats to `16/00-0012345C16` (*Figure 11*).	🟢 PASS
04	`S04`	**RC Legacy Wilaya 00 (National)**	`00B0012637`	Grandfathered registry entity (BELLAT FER DISTRIBUTION) formatted as `00/00-0012637B00` (*Figure 9*).	🟢 PASS
05	`S05`	**15-Digit Corporate NIF (Prefix 0)**	`002025007288530`	Validates standard 15-digit company fiscal number, enriches legal card, sets `is_company=True`.	🟢 PASS
06	`S06`	**Legacy Wilaya Prefix NIF**	`000816097764131`	Pre-2022 company with legacy Wilaya `00` / `000` prefix accepted and resolved without truncation.	🟢 PASS
07	`S07`	**15-Digit Natural Person NIF (Prefix 1)**	`198451010000284`	Classifies contact as `Personne Physique` (`is_company=False`) in compliance with Law 18-07.	🟢 PASS
08	`S08`	**NIS Identifier Resolution**	`002025007288530`	`_onchange_kyb_nis` triggers gateway lookup and resolves counterparty NIF and official RC.	🟢 PASS
09	`S09`	**Real-Time Name Autocomplete**	Query: `DIGI DEV`	Intercepts `_name_search`, returns live candidate dropdown matches from national registry (*Figure 12*).	🟢 PASS
10	`S10`	**Autocomplete Local Existence Badge**	Query: `DIGI DEV`	Tags existing local database contacts with `• ⚠️ Existant (#ID)` to alert users before creation.	🟢 PASS
11	`S11`	**Duplicate NIF Policy `warn`**	NIF: `002025007288530`	On duplicate NIF entry, displays institutional warning dialog referencing existing contact (*Figure 13*).	🟢 PASS
12	`S12`	**Duplicate NIF Policy `block`**	NIF: `002025007288530`	When policy set to `block`, raises `UserError` rejecting creation or modification of duplicate NIF.	🟢 PASS
13	`S13`	**Non-Existent / Unindexed Query**	`NonExistentXYZ99999`	Gracefully handles queries not found in registry; surfaces non-blocking warning toast with zero crashes.	🟢 PASS

#	Code	Scenario Description	Tested Identifier / Action	Verified Behavior & Screenshot Reference	Status
14	`S14`	**Invoice with Active Vendor**	Date: `2026-10-10`	Evaluates vendor standing at invoice date: status `compliant`, `tax_deductible=True` (*Figure 15 & 16*).	🟢 PASS
15	`S15`	**Invoice with Radié Vendor (Post-Cutoff)**	Date: `2026-10-10`	Vendor radié in 2021: marks invoice `non_compliant`, `tax_deductible=False` (*Figure 17 & 19*).	🟢 PASS
16	`S16`	**Strict Invoicing Blocking Policy**	Action: `action_post`	When policy is `strict_block`, prevents accounting confirmation of non-compliant vendor bill.	🟢 PASS
17	`S17`	**Cautionary Historical Invoice**	Date: `2020-01-15`	Invoice dated *prior* to vendor's 2021 radiation date: evaluates to cautionary `warning` status (*Figure 18*).	🟢 PASS
18	`S18`	**Native Odoo Deduplication Wizard**	`action_merge`	Merging duplicate partners consolidates cards while preserving all canonical KYB audit fields (*Figure 14*).	🟢 PASS

9. Sandbox Testing Matrix (Deterministic Magic Keys)

For staging environments and automated CI/CD pipelines, KYB provides deterministic mock fixtures that bypass billing and external scraping:

Test Identifier	Category	Simulated Standing	Result / Behavior
`000000000000001` / `16B0000001`	Personne Morale	**ACTIF**	Resolves SARL ALGERIE TECH (`ACTIF`).
`000000000000002` / `00B0000002`	Personne Morale	**RADIÉ**	Resolves SARL ANCIENNE DISSOLTE (Radié 15/06/2021).
`51/00-0000006A` / `51A0000006`	Personne Physique	**ACTIF**	Resolves individual merchant (`is_company=False`).
`000000000000404` / `404RC`	Any	**NOT FOUND**	Returns HTTP 404 `ENTITY_NOT_FOUND`.
`000000000000402`	Any	**QUOTA**	Returns HTTP 402 `QUOTA_EXHAUSTED`.

10. Troubleshooting & FAQ

1. **Why does autocomplete not return natural person names?**

Under **Algerian Law 18-07** and statutory corporate privacy standards, free-text name discovery is strictly restricted to Personnes Morales (companies). Natural persons (Personnes Physiques) must be resolved deterministically using their official **Category A/C RC** or **15-digit NIF**.

1. **What should I do if an invoice is flagged "Fournisseur Radié"?**

Inspect the `kyb_date_radiation` on the vendor card. If the invoice date postdates the strike-off date, verify the transaction with your tax accounting department under Articles 169 & 176 du CIDTA.

1. **How do I merge duplicate partner records?**

Go to **Contacts**, select the duplicate records in list view, and click **Actions** → **Fusionner**. Select the canonical partner as destination. The KYB module automatically preserves all fiscal audit data.

1. **How do I rotate an API Sub-Key if an employee leaves?**

Log in to the KYB Customer Portal, locate the Sub-Key assigned to your Odoo environment, click **Revoke**, and generate a new Sub-Key. Paste the new key into Odoo's KYB Settings panel.
